

Supplementary Information: Physics World Models for Computational Imaging

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1 Supplementary Note 1: Triad Decomposition Mathematical Derivations

The Triad Decomposition formalizes three successive gates that every computational-imaging reconstruction must pass. We derive rigorous bounds for each gate and show how the *recovery ratio* ρ summarizes the overall reconstruction quality.

1.1 Gate 1 — Information-Theoretic Limit (Compression)

Let the forward operator $\mathbf{H} \in \mathbb{R}^{m \times n}$ map an n -dimensional scene $\mathbf{x}$ to m measurements $\mathbf{y}$. Define the *compression ratio* $\gamma = m/n$.

Theorem 1 (Compression bound). *For any estimator $\hat{\mathbf{x}}(\mathbf{y})$, the minimum achievable mean-squared error satisfies*

$$\text{MSE}_{\min} \geq \frac{1}{n} \sum_{i=1}^{n-m} \sigma_i^2(\mathbf{x}), \quad (\text{S1})$$

where $\sigma_i^2(\mathbf{x})$ denotes the variance of $\mathbf{x}$ along the i -th null-space direction of $\mathbf{H}$.

Interpretation. The null space of $\mathbf{H}$ has dimension $\dim \mathcal{N}(\mathbf{H}) = n - \text{rank}(\mathbf{H}) \geq n - m$. Information lost in the null space cannot be recovered without a prior; Gate 1 therefore sets an *information-theoretic floor* on PSNR:

$$\text{PSNR}_{\max}^{(G1)} = 10 \log_{10} \left(\frac{\|\mathbf{x}\|_{\infty}^2}{\text{MSE}_{\min}} \right). \quad (\text{S2})$$

Modalities with higher compression ratios (e.g. CACTI with B frames compressed into one) face a proportionally lower ceiling.

1.2 Gate 2 — SNR-Dependent PSNR Bound (Noise)

Given additive noise $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I})$, the measurement SNR is $\text{SNR} = \|\mathbf{H}\mathbf{x}\|^2 / (m \sigma_n^2)$.

Theorem 2 (Noise bound). *Under matched forward model and Gaussian noise, the best achievable PSNR is*

$$\text{PSNR}_{\max}^{(G2)} = 10 \log_{10}(\text{SNR}) + C_{\mathcal{M}}, \quad (\text{S3})$$

where $C_{\mathcal{M}}$ is a modality-dependent constant that absorbs the conditioning of $\mathbf{H}$ and the prior strength:

$$C_{\mathcal{M}} = 10 \log_{10} \left(\frac{n \|\mathbf{x}\|_{\infty}^2}{\|\mathbf{x}\|^2} \cdot \kappa^{-2}(\mathbf{H}) \right), \quad (\text{S4})$$

with $\kappa(\mathbf{H})$ the condition number of $\mathbf{H}$ restricted to its column space.

Interpretation. Gate 2 is a *noise ceiling*: no algorithm can exceed $\text{PSNR}_{\max}^{(G2)}$ regardless of computational budget. Empirically, modalities such as MRI ($C_{\text{MRI}} \approx 8$ dB) are more noise-tolerant than CASSI ($C_{\text{CASSI}} \approx 3$ dB) due to favorable operator conditioning.

1.3 Gate 3 — Sensitivity to Model Mismatch (Calibration)

Let $\boldsymbol{\theta} \in \mathbb{R}^p$ collect the calibration parameters (e.g. mask shift, PSF blur, coil sensitivity) and let $\mathbf{H}_{\boldsymbol{\theta}}$ denote the parameterized forward operator.

Theorem 3 (Calibration sensitivity). *For small perturbation $\delta\boldsymbol{\theta}$ around the true parameter $\boldsymbol{\theta}^*$, the PSNR degradation is*

$$\Delta\text{PSNR} \approx -\frac{10}{\ln 10} \frac{\delta\boldsymbol{\theta}^\top \mathbf{J}^\top \mathbf{J} \delta\boldsymbol{\theta}}{\text{MSE}_0}, \quad (\text{S5})$$

where $\mathbf{J} = \partial(\mathbf{H}_{\boldsymbol{\theta}}\mathbf{x})/\partial\boldsymbol{\theta}|_{\boldsymbol{\theta}^*}$ is the parameter Jacobian and MSE_0 is the noise-only MSE at $\boldsymbol{\theta}^*$.

Per-parameter sensitivity. Restricting to a single parameter θ_j :

$$\frac{d\text{PSNR}}{d\theta_j} = -\frac{10}{\ln 10} \frac{\|\mathbf{J}_{:,j}\|^2}{\text{MSE}_0} \delta\theta_j + \mathcal{O}(\delta\theta_j^2). \quad (\text{S6})$$

The first-order Taylor expansion shows that PSNR degrades linearly in $|\delta\theta_j|$ for small mismatches and quadratically for larger ones.

1.4 Recovery Ratio

We define the *recovery ratio* as

$$\rho = \frac{\text{PSNR}_{\text{achieved}} - \text{PSNR}_{\text{mismatch}}}{\text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{mismatch}}}, \quad 0 \leq \rho \leq 1. \quad (\text{S7})$$

In rare cases where Scenario III exceeds Scenario I (e.g., due to regularization benefits from the corrected operator), ρ may exceed 1.

Proposition 4. *Under convex reconstruction loss with matched regularization and convex constraint on $\boldsymbol{\theta}$, the recovery ratio satisfies $0 \leq \rho \leq 1$ for any estimator.*

Proof sketch. The oracle PSNR is the global optimum of the jointly convex problem in $(\mathbf{x}, \boldsymbol{\theta})$. The mismatch PSNR uses a fixed $\boldsymbol{\theta}_0 \neq \boldsymbol{\theta}^*$, yielding a suboptimal feasible point. Any correction step moves towards the optimum, so $\text{PSNR}_{\text{mismatch}} \leq \text{PSNR}_{\text{achieved}} \leq \text{PSNR}_{\text{oracle}}$, giving $\rho \in [0, 1]$. $\square$

Under the conditions of Proposition 1, $0 \leq \rho \leq 1$. In practice, beneficial regularization bias from the corrected operator can yield $\rho > 1$, as observed for CACTI.

2 Supplementary Note 2: Complete OperatorGraph Specification

The `OperatorGraph` is the central abstraction of the PWM framework. It represents a computational imaging pipeline as a directed acyclic graph (DAG) of differentiable operators.

2.1 Node Interface

Every node v in the graph implements the following interface:

Method	Contract
<code>forward(x) → y</code>	Apply the physical forward model. Input shape: <code>(B, C, *spatial)</code> . Output shape determined by the operator (e.g. compression changes spatial dims).
<code>adjoint(y) → x</code>	Apply the adjoint (transpose) operator. Must satisfy $\langle \mathbf{H}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{H}^\top \mathbf{y} \rangle$ to numerical precision ($< 10^{-6}$ relative error).
<code>shape_in / shape_out</code>	Static shape annotations for compile-time validation.
<code>dtype</code>	Data type contract: <code>float32</code> (default) or <code>complex64</code> .
<code>parameters() → dict</code>	Returns learnable calibration parameters with their current values.

2.2 Edge Semantics

An edge $e = (u, v)$ represents data flow: the output tensor of node u is fed as input to node v . Edges carry the following metadata:

- **Tensor shape:** inferred at compile time via shape propagation.
- **Data type:** must match between source output and target input.
- **Optional transform:** lightweight reshape or permute operations (e.g. channel stacking for multi-frame CACTI data).

2.3 Compilation Algorithm

Given an `OperatorGraph` $G = (V, E)$:

1. **Topological sort.** Compute a valid execution order $v_1, v_2, \dots, v_{|V|}$ using Kahn’s algorithm. Raise `CycleError` if the graph contains a cycle.
2. **Shape propagation.** Starting from the source node(s), propagate tensor shapes through each operator’s `shape_out` to validate all edge contracts.
3. **Automatic adjoint chain.** Construct the adjoint graph $G^\top$ by reversing all edges and replacing each node’s `forward` with `adjoint`. The adjoint graph is used for gradient-based reconstruction and the Triad Decomposition sensitivity analysis.

4. **Fusion (optional).** Consecutive linear operators are fused into a single matrix–vector product when dimensions permit, reducing memory traffic.

2.4 JSON Serialization Schema

The graph is serialized to a JSON document for reproducibility and sharing:

```
{
  "version": "0.3.0",
  "nodes": [
    {
      "id": "mask",
      "type": "CassiMask",
      "params": {"shift_r": 0.0, "shift_c": 0.0},
      "shape_in": [1, 28, 256, 256],
      "shape_out": [1, 1, 256, 310]
    }
  ],
  "edges": [
    {"src": "mask", "dst": "detector", "dtype": "float32"}
  ],
  "metadata": {
    "modality": "CASSI",
    "created": "2026-02-01T00:00:00Z"
  }
}
```

3 Supplementary Table S1: All 16 Correction Configurations

Table S1 reports the correction results for all 16 correction configurations across the validated modalities. The first 9 modalities have full end-to-end validation; the remaining 7 are designated Phase 2/4 additions with partial or planned validation.

Table S1: Complete correction parameter summary. Δ PSNR is the improvement from Scenario II (mismatch) to Scenario III (corrected). CASSI correction results in this table use GAP-TV as the reconstruction solver; the main-text deep dive reports results for GAP-TV, MST-L, and HDNet. Scenario I (ideal) PSNR values and recovery ratios (ρ) for all configurations are available in the RunBundle manifests in the code repository.

Modality	Mismatch Parameter	Sc. II (dB)	Sc. III (dB)	Δ PSNR (dB)	RMSE
Matrix ¹	gain_bias	11.14	23.35	+12.21	0.0042
CT	center_of_rotation	13.41	24.09	+10.68	0.0031
CACTI	mask_timing	14.48	37.42	+22.94	0.0018
Lensless	psf_shift	23.48	27.03	+3.55	0.0089
MRI	coil_sensitivities	6.94	55.19	+48.25	0.0003
SPC	gain_bias	11.14	23.35	+12.21	0.0042
CASSI	mask_geo +	20.96	21.50	+0.54	0.0105
(Alg 1) ²	dispersion				
CASSI	mask_geo +	20.96	21.72	+0.76	0.0098
(Alg 2) ³	dispersion				
Ptychography	position_offset	17.35	24.44	+7.09	0.0063
<i>Phase 2/4 additions — validation in progress</i>					
Holography	propagation_distance	—	—	—	—
FPM	LED_positions	—	—	—	—
Phase	support_constraint	—	—	—	—
Retrieval					
OCT	dispersion_coeffs	—	—	—	—
Fluorescence	PSF_aberration	—	—	—	—
Ultrasound	speed_of_sound	—	—	—	—
Radar/SAR	motion_compensation	—	—	—	—

¹Matrix (Generic) is a canonical test configuration that uses the same gain-bias operator template as SPC; identical results confirm pipeline consistency rather than representing an independent modality.

²Reported values are averaged over 10 test scenes using GAP-TV as the reconstruction solver. Sc. II from InverseNet validation; Sc. III values are oracle upper bounds (true operator applied to mismatched data).

³Note: the correction-table PSNR reflects GAP-TV reconstruction; state-of-the-art solvers (e.g., MST-L) achieve 27.33 dB oracle PSNR under the corrected operator (see main text).

4 Supplementary Table S2: CASSI Per-Scene Results

Table S2 reports PSNR (dB) for 10 KAIST scenes across five representative methods under four scenarios.⁴

Table S2: CASSI per-scene PSNR (dB). Sc. I = ideal forward model; Sc. II = mismatched (baseline); Sc. III-A1 = correction Algorithm 1; Verify = pipeline sanity check (ideal-condition data with ideal operator, confirming pipeline integrity to ± 0.01 dB from Sc. I). GAP-TV values for Sc. I, Sc. II, and Sc. III-A1 are from InverseNet validation¹ under the 5-parameter mismatch ($dx=0.5$ px, $dy=0.3$ px, $\theta=0.1^\circ$, $a_1=2.02$, $\alpha=0.15^\circ$). Sc. III-A1 entries marked * use Oracle Mask values (true operator applied to mismatched data) as correction upper bounds. Algorithm 2 results are reported in Table S1.

Scene	Method	Sc. I	Sc. II	Sc. III-A1	Verify
scene01	GAP-TV	26.49	24.08	24.18*	26.49
scene01	TwIST	27.18	15.79	21.34	27.18
scene01	DeSCI	28.01	14.92	20.15	28.01
scene01	MST	33.47	16.84	22.61	33.47
scene01	HDNet	34.12	17.02	23.10	34.12
scene02	GAP-TV	24.60	21.89	22.82*	24.60
scene02	TwIST	25.73	15.21	20.48	25.73
scene02	DeSCI	26.95	14.38	19.54	26.95
scene02	MST	31.84	16.12	21.73	31.84
scene02	HDNet	32.56	16.45	22.01	32.56
scene03	GAP-TV	25.96	18.62	19.68*	25.96
scene03	TwIST	28.04	16.35	21.87	28.04
scene03	DeSCI	29.22	15.64	20.93	29.22
scene03	MST	34.67	17.48	23.25	34.67
scene03	HDNet	35.31	17.71	23.68	35.31
scene04	GAP-TV	28.36	23.37	24.41*	28.36
scene04	TwIST	31.09	18.02	23.75	31.09
scene04	DeSCI	32.14	17.11	22.64	32.14
scene04	MST	37.82	19.33	25.47	37.82
scene04	HDNet	38.45	19.68	25.94	38.45
scene05	GAP-TV	23.66	20.39	21.27*	23.66
scene05	TwIST	26.54	15.53	20.84	26.54
scene05	DeSCI	27.68	14.72	19.87	27.68
scene05	MST	32.91	16.67	22.14	32.91
scene05	HDNet	33.58	16.94	22.53	33.58
scene06	GAP-TV	22.34	20.39	21.00*	22.34
scene06	TwIST	29.18	16.74	22.31	29.18
scene06	DeSCI	30.42	15.97	21.24	30.42
scene06	MST	35.89	18.05	23.87	35.89
scene06	HDNet	36.54	18.32	24.25	36.54
scene07	GAP-TV	23.51	20.56	21.07*	23.51
scene07	TwIST	24.76	14.32	19.64	24.76
scene07	DeSCI	25.84	13.55	18.73	25.84
scene07	MST	30.72	15.48	21.02	30.72

⁴This supplementary table reports five methods (GAP-TV, TwIST, DeSCI, MST, HDNet) to provide comprehensive coverage; the main text focuses on three methods (GAP-TV, MST-L, HDNet) for clarity. The supplementary table uses MST (base variant); the main text reports MST-L (large variant), which has higher Scenario I performance but similar Scenario II collapse.

Scene	Method	Sc. I	Sc. II	Sc. III-A1	Verify
scene07	HDNet	31.35	15.73	21.38	31.35
scene08	GAP-TV	22.16	20.57	21.10*	22.16
scene08	TwIST	28.53	16.52	22.04	28.53
scene08	DeSCI	29.71	15.78	21.12	29.71
scene08	MST	35.14	17.83	23.55	35.14
scene08	HDNet	35.79	18.08	23.94	35.79
scene09	GAP-TV	23.03	19.14	20.61*	23.03
scene09	TwIST	26.89	15.64	21.02	26.89
scene09	DeSCI	28.04	14.91	20.08	28.04
scene09	MST	33.28	16.87	22.35	33.28
scene09	HDNet	33.94	17.12	22.71	33.94
scene10	GAP-TV	23.27	20.58	21.04*	23.27
scene10	TwIST	30.34	17.54	23.21	30.34
scene10	DeSCI	31.52	16.73	22.14	31.52
scene10	MST	37.05	18.89	24.91	37.05
scene10	HDNet	37.68	19.15	25.34	37.68

5 Supplementary Table S3: Full 26-Modality Benchmark

Table S3 lists all 26 modalities in the PWM benchmark, grouped by physical carrier tier.

Table S3: Full 26-modality benchmark. PSNR values are representative (Scenario I, default solver). “Ref. PSNR” is the best published result where available.

#	Modality	Domain	PSNR (dB)	Ref. PSNR (dB)	Status	Solver
1	Matrix (toy)	Incoherent	23.35	—	Validated	ADMM-TV
2	SPC	Incoherent	23.35	24.1	Validated	ADMM-TV
3	CASSI	Incoherent	24.34	34.2	Validated	GAP-TV
4	CACTI	Incoherent	35.33	33.4	Validated	EfficientSCI
5	Lensless	Incoherent	27.03	28.5	Validated	ADMM-TV
6	Fluorescence	Incoherent	—	30.1	Phase 4	—
7	Holography	Coherent	—	32.7	Phase 2	—
8	Ptychography	Coherent	24.44	26.8	Validated	ePIE
9	FPM	Coherent	—	31.5	Phase 2	—
10	Phase Retrieval	Coherent	—	29.3	Phase 2	—
11	CDI	Coherent	—	27.4	Phase 4	—
12	OCT	Coherent	—	35.2	Phase 4	—
13	CT (parallel)	X-ray	24.09	42.1	Validated	FBP + TV
14	CT (fan-beam)	X-ray	—	40.8	Phase 2	—
15	CT (cone-beam)	X-ray	—	39.5	Phase 2	—
16	Electron Pty.	Electron	—	25.9	Phase 4	—
17	MRI (Cartesian)	RF	55.19	42.3	Validated	CG-SENSE
18	MRI (radial)	RF	—	38.7	Phase 2	—
19	MRI (spiral)	RF	—	37.2	Phase 4	—
20	Ultrasound	Acoustic	—	31.8	Phase 4	—
21	Radar / SAR	RF	—	28.4	Phase 4	—
22	NeRF	Multi-view	—	31.7	Phase 2	—
23	3DGS	Multi-view	—	33.2	Phase 2	—
24	Light Field	Multi-view	—	36.1	Phase 4	—
25	Structured Illum.	Incoherent	—	32.4	Phase 4	—
26	Ghost Imaging	Incoherent	—	22.8	Phase 4	—

6 Supplementary Table S4: YAML Registry Summary

Table S4 summarizes the nine YAML registry files (totalling 7,034 lines) that define the PWM framework’s modular architecture.

Table S4: YAML registry files in `packages/pwm_core/contrib/`.

#	File	Lines	Purpose
1	<code>modalities.yaml</code>	1200	Defines 64 modality entries with keywords, forward model equations, and default solvers.
2	<code>graph_templates.yaml</code>	1400	OperatorGraph skeletons for 89 templates across 64 modalities.
3	<code>photon_db.yaml</code>	624	Photon models parameterized by source power, quantum efficiency, exposure, and detector.
4	<code>mismatch_db.yaml</code>	797	Mismatch parameters, perturbation ranges, and correction methods per modality.
5	<code>compression_db.yaml</code>	1186	Recoverability tables mapping compression ratio to expected PSNR with provenance.
6	<code>solver_registry.yaml</code>	650	Maps solver names to implementations (ADMM ² , FISTA ³ , GAP-TV, PnP, learned).
7	<code>primitives.yaml</code>	450	Primitive operator metadata (type, adjoint availability, differentiability).
8	<code>dataset_registry.yaml</code>	380	Links modalities to benchmark datasets (KAIST, fastMRI, etc.).
9	<code>acceptance_thresholds.yaml</code>	347	Pass/fail thresholds per metric for automated validation.

7 Supplementary Note 4: RunBundle Schema

The `RunBundle` is PWM’s reproducibility artifact. Every experiment produces a `RunBundle` that captures the complete computational state.

7.1 RunBundle v0.3.0 Specification

Table S5: RunBundle v0.3.0 field specification.

Field	Type	Description
<code>version</code>	string	Schema version, currently "0.3.0".
<code>git_hash</code>	string	Full 40-character SHA-1 of the commit used to generate results.
<code>rng_seeds</code>	object	Random seeds for Python (<code>random</code> , <code>numpy</code>), PyTorch (CPU, CUDA), and any solver-specific RNG.
<code>platform_info</code>	object	OS, Python version, GPU model, CUDA version, PyTorch version, and <code>pip freeze</code> hash.
<code>sha256_hashes</code>	object	SHA-256 checksums of all input data files (measurements, masks, ground truth).
<code>metrics</code>	object	Computed quality metrics: PSNR, SSIM, LPIPS, and any modality-specific metrics.
<code>timestamps</code>	object	ISO-8601 timestamps for start, end, and per-stage completion.
<code>operator_state</code>	object	Serialized <code>OperatorGraph</code> including all node parameters and edge metadata.
<code>triad_report</code>	object	Triad Decomposition evaluation: gate pass/fail status, PSNR bounds, and recovery ratio ρ .
<code>correction_trajectory</code>	array	Time series of correction iterations: <code>[{iter, params, loss, psnr, grad_norm}, ...]</code> . Enables convergence analysis and early-stopping diagnostics.
<code>solver_config</code>	object	Complete solver hyperparameters (algorithm, iterations, step size, regularization weight, denoiser checkpoint).
<code>modality</code>	string	Modality identifier matching <code>modalities.yaml</code> .
<code>scenario</code>	string	One of I (ideal), II (mismatch), III (corrected), IV (oracle_mask).
<code>notes</code>	string	Free-form text field for experiment annotations.

7.2 Integrity Verification

A `RunBundle` can be verified by:

1. Checking `git_hash` matches the repository state.
2. Re-computing `sha256_hashes` from the data files.
3. Re-running the solver with the stored `rng_seeds` and `solver_config` and comparing metrics within tolerance (± 0.01 dB PSNR).

8 Supplementary Note 5: Computational Cost Analysis

8.1 Runtime per Modality

All timings were measured on a single NVIDIA RTX 4090 GPU (24 GB VRAM) with PyTorch 2.1 and CUDA 12.1.

Table S6: Computational cost for correction (Scenario II $\rightarrow$ III). RoIC = dB recovered per GPU-hour.

Modality	Iterations	Runtime (s)	Δ PSNR (dB)	RoIC (dB/GPU-hr)
Matrix	200	12	+12.21	3663.0
CT	500	85	+10.68	452.3
CACTI	1000	180	+22.94	458.8
Lensless	300	45	+3.55	284.0
MRI	150	30	+48.25	5790.0
SPC	200	15	+12.21	2930.4
CASSI (Alg 1)	500	300	+0.54	6.5
CASSI (Alg 2)	2000	3200	+0.76	0.9
Ptychography	800	420	+7.09	60.8

8.2 RoIC Metric: Efficiency of Correction

We define the *Return on Invested Compute* (RoIC) as:

$$\text{RoIC} = \frac{\Delta\text{PSNR (dB)}}{T_{\text{GPU (hours)}}}, \quad (\text{S8})$$

where T_{GPU} is the wall-clock GPU time.

Key observations:

- **MRI** achieves the highest RoIC (5790 dB/GPU-hr) because coil sensitivity correction yields massive PSNR gains with a well-conditioned linear operator that converges rapidly.
- **Matrix/SPC** also show high RoIC due to the simplicity of the gain-bias correction model.
- **CASSI Algorithm 2** has the lowest RoIC (0.9 dB/GPU-hr) because it jointly optimizes mask geometry and dispersion over 2000 iterations, each requiring a full forward-adjoint cycle through the dispersive operator. Under the multi-parameter InverseNet mismatch, the absolute correction gains are modest (+0.76 dB), reflecting the inherent difficulty of simultaneously correcting five coupled mismatch parameters.
- The practical implication is that Algorithm 1 (fixed dispersion, correct mask geometry only) is preferred when compute budget is limited, recovering 71% of Algorithm 2’s Δ PSNR at <10% of the cost.

8.3 Scaling Considerations

For large-scale deployment:

- Correction is *embarrassingly parallel* across scenes/slices — multi-GPU scaling is near-linear.

- The OperatorGraph compilation overhead is amortized: once compiled, the same graph serves all scenes of a given modality.
- Memory footprint scales with the measurement dimension m , not the scene dimension n , making correction feasible even for high-resolution 3D volumes (e.g. 512^3 CT).

9 Supplementary Note 6: Real-Data Validation Details

9.1 CASSI Real Data: TSA Hyperspectral Camera

The TSA (Thermal Snapshot Aperture) real dataset consists of 5 hyperspectral scenes acquired with a coded aperture snapshot spectral imaging system. Each scene has spatial resolution 660×660 with 28 spectral bands (450–650 nm) and a mask-shift step of 2 pixels per band. The extended measurement has width $660 + (28 - 1) \times 2 = 714$ pixels.

Experimental protocol. For each scene, we reconstruct with four methods: GAP-TV (classical iterative), HDNet (mask-oblivious neural network), MST-S, and MST-L (mask-guided spectral transformers). Each method is applied under two conditions: (i) the calibrated mask provided with the dataset, and (ii) a perturbed mask with sub-pixel shift $dx = 0.5$ px, $dy = 0.3$ px. Because the MST model has a hardcoded 256×256 spatial assumption, the 660×660 real data is center-cropped to 256×256 for MST reconstruction. HDNet is fully convolutional and processes the full spatial extent.

Metric. Because no ground-truth hyperspectral cube exists for the real scenes, we use the measurement residual as a ground-truth-free diagnostic:

$$r = \frac{\|\mathbf{y} - H\hat{\mathbf{x}}\|^2}{\|\mathbf{y}\|^2}, \quad (\text{S9})$$

where $\mathbf{y}$ is the measured snapshot and $\hat{\mathbf{x}}$ is the reconstruction. The residual ratio $r_{\text{mismatch}}/r_{\text{calibrated}}$ quantifies the relative impact of operator mismatch, independent of the (unknown) ground truth.

Key findings. GAP-TV, which explicitly conditions on the coded aperture mask in its forward–adjoint iterations, shows the expected sensitivity to mask mismatch (mean ratio $1.8\times$). HDNet, whose architecture processes the measurement without explicit mask conditioning, is entirely insensitive (ratio $1.0\times$). The transformer methods (MST-S, MST-L) show ratios near or below $1.0\times$ on real data—in sharp contrast to their severe degradation in simulation (-13.98 dB for MST-L). This simulation-to-hardware gap indicates that the real hardware mask already contains uncorrected manufacturing imperfections, so the additional 0.5 px perturbation is small relative to pre-existing errors.

9.2 CACTI Real Data: Temporal Compressive Camera

The CACTI real dataset consists of 4 dynamic scenes (duomino, hand, pendulumBall, waterBallon) acquired with a coded aperture compressive temporal imaging system at 512×512 spatial resolution with compression ratio 10 (10 video frames encoded per snapshot).

Key findings. GAP-TV shows an order-of-magnitude residual increase ($10.4\times$ mean) under sub-pixel mask mismatch, consistent with the multiplicative error amplification in temporal compression (a single mask error propagates across all 10 compressed frames). PnP-FFDNet shows moderate robustness ($2.0\times$), likely because the FFDNet denoiser provides an implicit regularization that partially absorbs the mismatch artefacts. The CACTI results confirm that temporal compressive modalities are substantially more sensitive to operator mismatch than spectral compressive modalities (CASSI), a distinction that would be invisible without the unified TRIAD DECOMPOSITION framework.

Table S7: Supplementary Table S5: CASSI real-data measurement residuals. Residual values r (Eq. S9) and residual ratio (mismatched/calibrated) across 5 TSA scenes and 4 methods.

Scene	Method	r_{cal}	r_{mis}	Ratio
Scene 1	GAP-TV	0.00148	0.00298	2.0
Scene 1	HDNet	0.18506	0.18506	1.0
Scene 1	MST-S	0.09953	0.10746	1.1
Scene 1	MST-L	0.10055	0.09544	0.9
Scene 2	GAP-TV	0.00210	0.00331	1.6
Scene 2	HDNet	0.14853	0.14853	1.0
Scene 2	MST-S	0.25357	0.23807	0.9
Scene 2	MST-L	0.25441	0.22811	0.9
Scene 3	GAP-TV	0.00204	0.00320	1.6
Scene 3	HDNet	0.18291	0.18291	1.0
Scene 3	MST-S	0.09399	0.09160	1.0
Scene 3	MST-L	0.10127	0.09085	0.9
Scene 4	GAP-TV	0.00148	0.00297	2.0
Scene 4	HDNet	0.14355	0.14355	1.0
Scene 4	MST-S	0.15586	0.14712	0.9
Scene 4	MST-L	0.17899	0.15893	0.9
Scene 5	GAP-TV	0.00233	0.00421	1.8
Scene 5	HDNet	0.11325	0.11325	1.0
Scene 5	MST-S	0.11995	0.12004	1.0
Scene 5	MST-L	0.13008	0.12208	0.9
Mean	GAP-TV	0.00189	0.00333	1.8
Mean	HDNet	0.15466	0.15466	1.0
Mean	MST-S	0.14458	0.14086	1.0
Mean	MST-L	0.15306	0.13908	0.9

Table S8: Supplementary Table S6: CACTI real-data measurement residuals and total variation. Residual ratio (mismatched/calibrated) and total variation (TV) for 4 real scenes and 2 methods.

Scene	Method	r_{cal}	r_{mis}	Ratio
duomino	GAP-TV	8.0×10^{-6}	8.5×10^{-5}	10.6
duomino	PnP-FFDNet	0.00198	0.00403	2.0
hand	GAP-TV	7.0×10^{-6}	7.7×10^{-5}	11.0
hand	PnP-FFDNet	0.00249	0.00705	2.8
pendulumBall	GAP-TV	3.7×10^{-5}	3.46×10^{-4}	9.4
pendulumBall	PnP-FFDNet	0.00919	0.01150	1.3
waterBalloon	GAP-TV	1.4×10^{-5}	1.47×10^{-4}	10.5
waterBalloon	PnP-FFDNet	0.00264	0.00493	1.9
Mean	GAP-TV	—	—	10.4
Mean	PnP-FFDNet	—	—	2.0

Table S9: Supplementary Table S7: Grid-search calibration results on simulated data with known ground truth. PSNR values in dB.

Modality	Sc. II	Sc. IV	Sc. III	Time (s)	Recovery
CASSI	21.95	23.31	23.69	1234.3	78%
CACTI	17.60	26.99	26.99	64.2	100%
SPC	25.47	25.47	25.43	217.5	0%

9.3 Autonomous Calibration on Real Data

CACTI achieves perfect calibration recovery because the mismatch manifold is low-dimensional (2D spatial shift) and the temporal compression provides high sensitivity to mask position. CASSI achieves 78% recovery; the residual gap reflects the higher-dimensional mismatch space (mask geometry plus dispersion) and pre-existing hardware imperfections. SPC gain-drift mismatch is not recoverable from the measurement residual alone, because a multiplicative gain bias scales both signal and noise uniformly, leaving the residual structure unchanged. This demonstrates a fundamental limitation of residual-based calibration for gain-type mismatches and motivates the development of alternative diagnostics for GATE 3 in gain-dominated systems.

10 Supplementary Table S8: SSIM Comparison Across Modalities

Table S10 reports the structural similarity index (SSIM) for all three validated photon-domain modalities under the three primary scenarios. SSIM complements PSNR by capturing perceptual quality, particularly structural distortions from operator mismatch.

Table S10: SSIM comparison across modalities and scenarios. Values are mean $\pm$ s.d. across test scenes/images.

Modality	Method	Sc. I	Sc. II	Sc. III
<i>CASSI (10 KAIST scenes, 28 spectral bands)</i>				
	GAP-TV	0.723 ± 0.088	0.612 ± 0.084	0.688 ± 0.083
	HDNet	0.970 ± 0.009	0.756 ± 0.074	0.756 ± 0.074
	MST-L	0.973 ± 0.009	0.744 ± 0.069	0.881 ± 0.035
<i>CACTI (6 benchmark videos, 8 temporal frames)</i>				
	GAP-TV	0.848 ± 0.083	0.305 ± 0.070	0.794 ± 0.063
	PnP-FFDNet	0.890 ± 0.060	0.216 ± 0.076	0.820 ± 0.054
	ELP-Unfolding	0.965 ± 0.013	0.308 ± 0.076	0.927 ± 0.017
	EfficientSCI	0.973 ± 0.012	0.303 ± 0.079	0.927 ± 0.017
<i>SPC (11 standard images, 25% compression)</i>				
	FISTA-TV	0.852 ± 0.046	0.586 ± 0.055	0.759 ± 0.036
	PnP-DRUNet	0.880 ± 0.047	0.402 ± 0.062	0.645 ± 0.050
	ISTA-Net	0.916 ± 0.028	0.584 ± 0.077	0.760 ± 0.056
	HATNet	0.847 ± 0.049	0.648 ± 0.091	0.807 ± 0.060

Key observations. The SSIM degradation pattern mirrors the PSNR findings: under Scenario II (mismatch), all methods collapse to a narrow quality range regardless of their Scenario I performance. In CASSI, the SSIM collapse is particularly striking: MST-L drops from 0.973 to 0.744, a perceptual quality loss that exceeds the improvement from a decade of solver development. In CACTI, mismatch produces near-random SSIM values (~ 0.3), confirming that the reconstructed video frames bear little structural resemblance to the ground truth under mismatch conditions.

11 Supplementary Table S9: CASSI Spectral Angle Mapper (SAM)

Table S11 reports the spectral angle mapper (SAM, in degrees) for CASSI, which measures spectral fidelity independently of intensity. Lower values indicate better spectral reconstruction.

Table S11: CASSI spectral angle mapper (SAM, degrees). Lower is better. Values are mean $\pm$ s.d. across 10 KAIST scenes.

Method	Sc. I	Sc. II	Sc. III
GAP-TV	16.66 ± 3.32	24.27 ± 2.92	25.97 ± 2.54
PnP-HSI-CNN	16.10 ± 3.25	23.73 ± 2.81	18.66 ± 2.75
HDNet	6.67 ± 0.96	17.03 ± 2.76	17.03 ± 2.76
MST-L	7.44 ± 1.24	23.92 ± 4.58	11.74 ± 1.32

Key observations. Spectral fidelity degrades severely under mismatch: MST-L’s SAM increases from 7.44° (near-perfect spectral reconstruction) to 23.92° (substantial spectral mixing). Oracle correction partially recovers spectral quality (11.74°), but the residual SAM error confirms that multi-parameter mismatch produces irreversible spectral distortions not fully correctable by mask-only correction. The SAM metric is particularly valuable for remote sensing applications where spectral accuracy is more important than spatial quality.

12 Supplementary Tables S10–S11: Gate 1 and Gate 2 Validation

The main text argues that Gate 3 (operator mismatch) is the dominant bottleneck under standard operating conditions. To demonstrate that Gate 1 (information deficiency) and Gate 2 (carrier budget) are real and measurable, we sweep the compression level and noise level across all seven validated modalities while keeping the forward model perfectly calibrated. All experiments use classical solvers (no deep-learned components) to ensure the results reflect information-theoretic limits rather than solver-specific behaviour.

12.1 Gate 1: Information Deficiency (Extreme Compression)

Table S12 reports reconstruction PSNR (dB) as the compression ratio / sampling rate / blur width is driven to extreme values. The forward model is ideal in every case; any PSNR loss is attributable solely to information lost in the measurement process.

Table S12: Gate 1 validation: PSNR (dB) under extreme compression for 7 modalities. “Best solver” reports the better of the two classical methods tested. Bold entries highlight the regime where PSNR falls below 15 dB (severe information loss).

Modality	Parameter	PSNR (dB) at sweep points				
		Nominal	Moderate	Severe	Extreme	Critical
CACTI	CR	25.9 (8)	24.1 (16)	22.7 (32)	20.6 (64)	—
CASSI	Transmittance	26.1 (50%)	27.7 (25%)	27.7 (10%)	27.6 (5%)	27.4 (2%)
SPC	CS ratio	28.3 (25%)	24.0 (10%)	21.2 (5%)	16.2 (2%)	14.4 (1%)
MRI	Sampling rate	28.8 (25%)	25.1 (10%)	24.3 (5%)	23.9 (2%)	—
CT	Projection angles	22.1 (180)	22.0 (90)	21.2 (30)	20.7 (10)	17.8 (5)
Lensless	Blur σ (px)	36.6 (1)	25.5 (3)	23.1 (5)	20.5 (10)	17.9 (20)
Ptychography	Scan positions	15.0 (16)	12.4 (9)	12.0 (4)	9.9 (1)	—

Key observations.

- **SPC** and **Lensless** show the most dramatic Gate 1 collapse. SPC PSNR drops 13.9 dB from 25% to 1% CS ratio; lensless drops 18.7 dB as the PSF blur widens from $\sigma=1$ to $\sigma=20$ px. In both cases the loss is irrecoverable: no solver can reconstruct information that was never measured.
- **CASSI** is a notable exception: reducing mask transmittance from 50% to 2% produces no degradation (mean PSNR actually increases by ~ 1.5 dB). This occurs because sparser coded apertures reduce spectral mixing in the multiplexed measurement, making the demixing problem easier for iterative solvers. The CASSI null space is effectively unchanged because each spectral band still contributes signal through the remaining open mask pixels.
- **CACTI** degrades monotonically with compression ratio ($25.9 \rightarrow 20.6$ dB over $8\times$ CR increase), consistent with the growing null space when fewer temporal measurements are available.
- **CT** shows SART outperforming FBP at sparse angles (21.2 dB vs. 15.8 dB at 10 angles), illustrating that iterative solvers can partially compensate for angular undersampling via their implicit prior, but ultimately both methods converge at ≤ 5 angles.

12.2 Gate 2: Carrier Budget (Noise Sweep)

Table S13 reports reconstruction PSNR as the photon budget or noise level is swept to extreme values while the compression is held at a well-posed nominal value and the forward model is ideal.

Table S13: Gate 2 validation: PSNR (dB) under increasing noise for 7 modalities. “Best solver” reports the better of the two classical methods tested. Noise parameter: photon count for Poisson-dominated modalities, Gaussian σ (fraction of signal range) for MRI and SPC. Bold entries indicate $\text{PSNR} < 15 \text{ dB}$.

Modality	Noise param.	PSNR (dB) at sweep points				
		Low noise		Moderate		Extreme
CACTI	Photon count	24.8 (10k)	23.3 (1k)	18.4 (100)	10.5 (10)	—
CASSI	Photon count	24.6 (10k)	22.3 (1k)	20.2 (100)	17.0 (10)	—
SPC	σ (frac.)	27.9 (0)	27.8 (.01)	25.3 (.05)	21.6 (.1)	13.5 (.3)
MRI	σ (frac.)	28.8 (0)	17.1 (.01)	12.8 (.05)	12.1 (.1)	11.0 (.3)
CT	Photon count	22.1 (100k)	21.9 (10k)	21.0 (1k)	18.3 (100)	—
Lensless	Photon count	40.9 (10k)	32.8 (1k)	23.1 (100)	13.6 (10)	—
Ptychography	Photon count	13.5 (10k)	10.3 (1k)	10.0 (100)	9.7 (10)	—

Key observations.

- Every modality exhibits monotonic PSNR degradation with increasing noise, confirming that Gate 2 failures are universal.
- **MRI** is the most noise-sensitive modality: adding just $\sigma=0.01$ Gaussian noise to k-space drops CS-wavelet PSNR from 28.8 dB to 17.1 dB, a 11.7 dB cliff. This is because MRI’s Fourier encoding concentrates signal energy in a few k-space samples, so additive noise corrupts the information-dense center region disproportionately.
- **Lensless** imaging shows the widest Gate 2 dynamic range: 40.9 dB at 10,000 photons to 13.6 dB at 10 photons, a 27.3 dB span. This reflects the well-conditioned forward operator (diffuser PSF), which faithfully propagates both signal and noise.
- **CT** is the most noise-robust modality (3.8 dB drop from 10^5 to 10^2 photons), because the many-angle sinogram provides substantial redundancy that averages out Poisson noise.
- The cliff-edge nature of Gate 2 is most dramatic in CACTI (14.3 dB drop from 10,000 to 10 photons) and SPC (14.4 dB drop from $\sigma=0$ to $\sigma=0.3$), where the compressed measurement offers no noise-averaging redundancy.

Comparison with Gate 3. Under standard operating conditions (adequate compression, adequate photon budget), the Gate 3 (mismatch) degradation reported in Supplementary Table S1 is typically 3–20 dB—comparable in magnitude to Gate 1 and Gate 2 degradation, but *correctable* by operator refinement. This is the key distinction: Gate 1 and Gate 2 losses are information-theoretic and irrecoverable, whereas Gate 3 losses are recoverable via calibration. The practical dominance of Gate 3 arises because modern instruments operate well above the Gate 1 and Gate 2 thresholds, leaving operator mismatch as the binding constraint.

13 Supplementary Note 7: Clinical CT Quality Assurance Validation

The TRIAD DECOMPOSITION framework has been translated to clinical CT quality assurance (QA) through the CT QC Copilot module, which maps the three gates to clinical failure modes and implements ACR CT accreditation standards.

13.1 Gate Mapping to Clinical Failure Modes

Table S14: Mapping of Triad Decomposition gates to clinical CT QC failure modes.

Gate	Research Interpretation	Clinical Interpretation
Gate 1	Information deficiency (null space, compression)	Protocol design inadequacy (insufficient projections, FOV)
Gate 2	Carrier budget (SNR, photon count)	Dose budget (noise floor vs. diagnostic need)
Gate 3	Operator mismatch (mask shift, PSF error)	Scanner calibration drift (HU drift, CoR offset, gain)

13.2 ACR Metric Validation

Nine ACR-aligned QC metrics are computed automatically from CT phantom (Gammex 464) scans. Table S15 reports the measurement agreement between the CT QC Copilot and console-reported values on a GE Revolution Apex scanner (120 kVp, 200 mAs, 5 mm axial).

Table S15: ACR metric agreement: CT QC Copilot vs. console gold standard.

Metric	Copilot	Console	Deviation	% Tolerance
CT# Water (HU)	0.3	0.2	0.1	2%
CT# Bone (HU)	952	953	1.0	5%
CT# Air (HU)	-997	-998	1.0	5%
CT# Acrylic (HU)	120	121	1.0	7%
CT# Polyethylene (HU)	-96	-96	0.0	0%
Geometric accuracy (mm)	0.15	0.10	0.05	2.5%
Slice thickness (mm)	5.08	5.02	0.06	4%
Uniformity (HU)	1.2	1.1	0.1	2%
Noise std. dev. (HU)	4.6	4.5	0.1	10%
Spatial resolution (lp/cm)	6.2	6.0	0.2	4%

All metrics are within $\leq 10\%$ of clinical tolerance bands, confirming that automated computation agrees with manual physicist measurements to within acceptable precision.

13.3 Drift Detection Performance

Statistical process control (SPC) using five Western Electric rules was evaluated on a 30-scanner simulated fleet over 12 monthly measurement cycles. Four scanners were programmed with known

drift trajectories (noise drift: 2 scanners; uniformity drift: 1 scanner; CT number drift: 1 scanner).

- **Sensitivity:** 100% (4/4 drifting scanners detected; 95% CI: 39.8%–100%)
- **Specificity:** 100% (0/26 false positives on stable scanners; 95% CI: 86.8%–100%)
- **Early warning lead time:** 3–6 months before ACR threshold exceedance

The automated detection provides a proactive maintenance window that is not achievable with manual threshold-based QA, which only triggers at the point of failure.

13.4 Workflow Efficiency

Table S16: Per-scanner QC workflow time comparison.

Stage	Manual (min)	Copilot (min)
DICOM ingestion	5.0	0.01
Metric computation	25.0	0.01
Threshold evaluation	5.0	<0.01
Trend analysis	10.0	<0.01
Report generation	15.0	0.02
Physicist review/sign-off	7.0	4.0
Total	67 ± 12	4.2 ± 0.8

The 94% reduction in per-scanner QC time (from 67 ± 12 to 4.2 ± 0.8 minutes) translates to 377 physicist-hours annually for a 30-scanner fleet (0.18 FTE reallocation). Bit-exact reproducibility (verified by SHA-256 comparison across 100 identical runs) eliminates inter-analyst variability, a known source of inconsistency in manual QA workflows.

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